

Fourth-order asymptotics and monotonicity in slope-constrained moment design

Hüseyin Buldurgan

Independent researcher

23 September 2026

Preprint prepared for mathematical review

Abstract

We study the least amplitude required to satisfy finitely many linear moment conditions under a Lipschitz constraint. Starting from a sign certificate for the unconstrained problem, we obtain a fourth-order expansion in the reciprocal slope budget. Its coefficient contains a nonnegative quadratic correction caused by the lower moment constraints. The result covers both finitely many switches on a compact interval and countably many separated switches on the half-line, under an explicit weighted summability condition. Examples show that the fourth-order coefficient can have either sign, and that twice continuous differentiability alone does not imply a fourth-order expansion. For a positive theta-kernel family along a short, validated cusp branch, we give uniform, explicit sixth-order error bounds. A weighted change of scale then proves strict ordering of the actual optimal values for every pair of distinct branch parameters, including arbitrarily close pairs. The numerical claims are accompanied by interval certificates and reproducible checks. This preprint is prepared for independent mathematical review; neither a formal verification nor an external referee report is claimed.

Keywords. Moment constraints; Lipschitz optimization; bounded Lipschitz norm; asymptotic expansion; computer-assisted proof; theta kernel; cusp.

1 Problem and contribution

Let X be a compact interval or $[0, \infty)$, let $q_0, \dots, q_m$ be real integrable functions, and let $m \geq 1$, $f > 0$. For $M > 0$, define

$$\delta(M) = \inf \left\{ \|g\|_\infty : \text{Lip}(g) \leq M, \quad \int_X q_j g = 0 \quad (0 \leq j < m), \quad \int_X q_m g = f \right\}. \quad (1.1)$$

The functions g are real and Lipschitz; endpoint values are free. Amplitude and slope are separate constraints. Put

$$q_b = q_m - \sum_{j < m} b_j q_j, \quad H_b(a) = \sup_{\|s\|_\infty \leq 1, \text{Lip}(s) \leq 1/a} \int_X q_b s.$$

The latter is a one-dimensional bounded Lipschitz support problem, closely related to the Kantorovich–Rubinstein and Fortet–Mourier frameworks [1, 2, 3]. We do not claim a new dual norm or a new principle of slope saturation.

Assume that a vector b^* gives the sign certificate

$$q_* = q_{b^*}, \quad \int_X q_j \text{sgn } q_* = 0 \quad (j < m), \quad D = \int_X |q_*| > 0. \quad (1.2)$$

Then the problem without the Lipschitz constraint has value $\delta_0 = f/D$. Indeed $f = \int q_* g \leq D\|g\|_\infty$, and $g = \delta_0 \operatorname{sgn} q_*$ realizes equality in the measurable class.

Our main result identifies both coefficients in

$$\delta(M) = \delta_0 + \frac{C_2}{M^2} + \frac{C_4}{M^4} + o(M^{-4}) \quad (M \rightarrow \infty).$$

The explicit fourth coefficient, the treatment of an infinite switch set, and the application with uniform error and transport bounds are the contributions for which this manuscript seeks review. We distinguish these claims from the established optimization framework. An exhaustive priority claim is not made.

Section 2 states and proves the general result. Section 3 gives exact examples and a regularity obstruction. Section 4 specializes it to a positive integral family and states the computer-assisted bounds. Section 5 proves a general transport comparison and its strict-order consequence. The appendices explain the effective estimates and the interface between the analytic proof and the computational evidence.

2 Fourth-order value asymptotics

At each simple zero z_k of q_* , write

$$Q_k = (q_j(z_k))_{j < m}, \quad r_k = q'_*(z_k), \quad t_k = q''_*(z_k), \quad u_k = q'''_*(z_k), \quad \gamma_k = |r_k|, \quad \sigma_k = \operatorname{sgn} r_k.$$

Define the following quantities whenever their series converge:

$$\Gamma = \sum_k \gamma_k, \quad G = 2 \sum_k \frac{Q_k Q_k^\top}{\gamma_k}, \quad (2.1)$$

$$B_j = \frac{1}{3} \sum_k \sigma_k \left(\frac{q_j(z_k) t_k}{r_k} - q'_j(z_k) \right), \quad v = G^{-1} B,$$

$$R = \sum_k \left(\frac{t_k^2}{36 \gamma_k} - \frac{\sigma_k u_k}{60} \right), \quad P = \frac{1}{2} B^\top G^{-1} B, \quad \Xi = R - P. \quad (2.2)$$

Here G is an m -by- m matrix and $B, v \in \mathbb{R}^m$.

2.1 Countably many switches

We impose the following hypotheses on $X = [0, \infty)$.

- (H1) The functions q_j are in $L^1(X)$ and are locally C^3 . The sign certificate (1.2) holds, $q_*(0) \neq 0$, and the positive zeros z_k tend to infinity with positive minimum separation.
- (H2) There are disjoint neighborhoods $N_k = [z_k - \rho, z_k + \rho] \subset (0, \infty)$, with a common $\rho > 0$, and a bounded open neighborhood U of b^* . For every $b \in U$, q_b has exactly one zero $z_k(b)$ in each N_k and no other zeros. The orientations remain σ_k . The functions $z_k(b)$ are C^1 , have a common Lipschitz constant, and $|z_k(b) - z_k| < \rho/4$.
- (H3) There are $L_k > 0$ such that $|q_b(x)| \geq L_k |x - z_k(b)|$ on N_k for all $b \in U$. Set

$$E_k = \sum_{j=0}^m \sum_{\ell=0}^3 \sup_{N_k} |q_j^{(\ell)}|, \quad C_U = 1 + \sup_{b \in U} \sum_{j < m} |b_j|, \quad K_k = \frac{C_U E_k}{6 L_k}.$$

The weighted condition is

$$\sum_k E_k (1 + K_k)^2 < \infty. \quad (2.3)$$

The sequence K_k need not be bounded.

(H4) Some m vectors among the Q_k are linearly independent.

Theorem 2.1. *Under (H1)–(H4), all the series in (2.1)–(2.2) converge absolutely, G is positive definite, and (1.1) has an optimizer for all sufficiently large M . Moreover,*

$$\delta(M) = \delta_0 + C_2 M^{-2} + C_4 M^{-4} + o(M^{-4}), \quad (2.4)$$

$$C_2 = \frac{\delta_0^3 \Gamma}{3D}, \quad C_4 = \delta_0^5 \left(\frac{\Gamma^2}{3D^2} - \frac{R}{D} + \frac{P}{D} \right). \quad (2.5)$$

The term P is nonnegative and is unchanged by any invertible linear change of basis of the lower moment functions $q_0, \dots, q_{m-1}$.

We give the proof, including the tail and exact-feasibility steps. For an average over $[-1, 1]$, write $\mathbb{E}h(y) = \frac{1}{2} \int_{-1}^1 h(y) dy$.

Lemma 2.2 (Balanced ramps). *For all $b \in U$ sufficiently close to b^* and all sufficiently small $a > 0$, there is a unique center $c_k \in (z_k(b) - a, z_k(b) + a)$ satisfying*

$$\mathbb{E}q_b(c_k + ay) = 0. \quad (2.6)$$

Its displacement satisfies

$$|c_k - z_k(b)| \leq \min\{a, K_k a^2\}.$$

The function equal to $\text{sgn } q_b$ outside the ramp intervals and equal to $\sigma_k(x - c_k)/a$ on $[c_k - a, c_k + a]$ maximizes $H_b(a)$.

Proof. At $c = z_k(b) - a$ and $c = z_k(b) + a$, the averaged function has opposite signs. Between them its derivative is $[q_b(c + a) - q_b(c - a)]/(2a)$, of strict sign σ_k . Taylor's theorem gives $|q_b(c) - \mathbb{E}q_b(c + ay)| \leq C_U E_k a^2/6$. The crossing bound then proves the displacement estimate.

On a balanced interval $[\ell, r]$, put $J(x) = \int_\ell^x q_b(t) dt$. There is just one change of sign of q_b , and $J(\ell) = J(r) = 0$, $\sigma_k J(x) < 0$ inside. For any competitor s , integration by parts gives

$$\int_\ell^r q_b(s - s_a) = - \int_\ell^r J(x)(s'(x) - \sigma_k/a) dx \leq 0.$$

Outside these intervals, $s_a = \text{sgn } q_b$ maximizes the integrand pointwise. The intervals are disjoint for small a . The infinite sum is justified by the integrable bound $2|q_b|$. $\square$

Proof of Theorem 2.1. Convergence and coefficients. Differentiating the zero equation gives $\partial_{b_j} z_k = q_j(z_k)/r_k$. The common root Lipschitz bound in (H2) therefore bounds Q_k/r_k . Also $|t_k/r_k| \leq 6K_k$. These estimates and (2.3) imply absolute convergence of Γ, G, B, R . Condition (H4) makes G positive definite.

Choose the trial multiplier $b(a) = b^* + va^2$ and its balanced ramp profile s_a . This trial multiplier is not asserted to solve the exact dual optimization problem. Define

$$\kappa_k = \frac{Q_k^\top v - t_k/6}{r_k}. \quad (2.7)$$

Implicit differentiation of the zero equation and (2.6), for each fixed k , give

$$\Delta_k := c_k - z_k = \kappa_k a^2 + o(a^2).$$

Uniformly in k , for a fixed constant C ,

$$|\Delta_k| \leq a + C\|v\|a^2 \leq 2a, \quad |\Delta_k|/a^2 \leq K_k + C\|v\|. \quad (2.8)$$

The first estimate and the second serve different purposes; we do not replace them by a false uniform bound on κ_k .

Summed moment expansions. For $H_j(t) = \int_t^\infty q_j(x) dx$, the moment change at a switch is

$$2\sigma_k\{\mathbb{E}H_j(z_k + \Delta_k + ay) - H_j(z_k)\}.$$

Only these differences are summed; separate sums of the $H_j(z_k)$ are unnecessary. If $h = \Delta_k + ay$, then

$$\mathbb{E}h = \Delta_k, \quad \mathbb{E}h^2 = \Delta_k^2 + a^2/3, \quad \mathbb{E}h^3 = \Delta_k^3 + \Delta_k a^2, \quad \mathbb{E}h^4 = \Delta_k^4 + 2\Delta_k^2 a^2 + a^4/5.$$

Taylor expansion and (2.8) now yield

$$\int q_j s_a = a^2(B - Gv)_j + o(a^2) = o(a^2) \quad (j < m), \quad (2.9)$$

$$\int q_* s_a = D - \frac{\Gamma a^2}{3} - a^4 \sum_k \sigma_k \left(r_k \kappa_k^2 + \frac{t_k \kappa_k}{3} + \frac{u_k}{60} \right) + o(a^4). \quad (2.10)$$

Here the normalized lower moment terms and the target terms are dominated by a constant times $E_k(1 + K_k)^2$. For example, $r_k \Delta_k^2/a^4$ uses the second bound in (2.8); terms of degree at least four use $|\Delta_k| \leq 2a$ as well. The Taylor remainder divided by a^4 tends to zero for each fixed k , by continuity of q_*''' , and has the same summable majorant. Dominated convergence proves both displayed expansions.

Completing the square with (2.7) gives

$$- \sum_k \sigma_k (r_k \kappa_k^2 + t_k \kappa_k/3 + u_k/60) = R - \frac{1}{2} v^\top G v = \Xi.$$

Since $q_{b(a)} = q_* - a^2 \sum v_j q_j$, (2.9) also gives the support expansion

$$H_{b(a)}(a) = D - \Gamma a^2/3 + \Xi a^4 + o(a^4). \quad (2.11)$$

For a general trial vector v , its fourth coefficient would be $R + \frac{1}{2} v^\top G v - B^\top v$; its minimum occurs at $G^{-1}B$.

Exact moment repair. Select the m independent switches from (H4), and move only their centers by a vector e . At $a = 0, e = 0$, the derivative of the lower moment map has columns $-2\sigma_k Q_k$, so it is invertible. For small a this derivative remains uniformly close to that invertible matrix. A contraction with that matrix as preconditioner, on a ball of radius a fixed multiple of the defect in (2.9), gives a continuous solution $e(a) = o(a^2)$. Indeed its forcing is $o(a^2)$, and its derivative norm can be made less than $1/2$, so that this ball maps into itself. Ramp intervals remain disjoint and the amplitude and slope bounds are unchanged. The derivative of the q_* moment with respect to each selected center is $O(a^2)$, while its second derivative is bounded. Consequently the repair changes that moment by $O(a^2|e| + |e|^2) = o(a^4)$. The repaired profile $\tilde{s}_a$ therefore has exact lower moments and

$$S(a) := \int q_m \tilde{s}_a = D - \Gamma a^2/3 + \Xi a^4 + o(a^4). \quad (2.12)$$

Put $A(a) = f/S(a)$ and $M(a) = A(a)/a$. The feasible function $A(a)\tilde{s}_a$ has amplitude $A(a)$ and slope at most $M(a)$. Both functions are continuous for small positive a , $A(a) \rightarrow \delta_0$, and $M(a) \rightarrow \infty$. The image of this interval under $M(a)$ is an interval unbounded above, hence contains all sufficiently large budgets. No monotonicity of this parametrization is needed.

Attainment and matching lower bound. For a fixed sufficiently large M , a minimizing sequence is uniformly bounded and equicontinuous. A diagonal Arzelà–Ascoli argument gives

a locally uniform limit. Dominated convergence against each $q_j \in L^1$ preserves the moments, and the norm and Lipschitz bounds pass to the limit. Thus the optimum is attained.

For any feasible amplitude A , use $a = A/M$, $s = g/A$, and the trial multiplier at that a . Then

$$f = \int q_{b(a)} g \leq AH_{b(a)}(a).$$

The preceding feasible upper bound and $\delta(M) \geq \delta_0$ show that the minimizing amplitude tends to δ_0 . The support and primal expansions therefore have matching scalar inversions. Explicitly, with $t = \Gamma/(3D)$, $e = \Xi/D$, the equation to this order is

$$A\{1 - t(A/M)^2 + e(A/M)^4 + o(M^{-4})\} = \delta_0.$$

Its increasing polynomial part has derivative bounded away from zero near δ_0 . The remainder is uniform for A in a compact neighborhood of δ_0 . Substitution yields

$$A = \delta_0 + t\delta_0^3 M^{-2} + (3t^2 - e)\delta_0^5 M^{-4} + o(M^{-4}),$$

which proves (2.5). In particular the factor 3 accounts for the dependence $a = A/M$. Finally, a lower-moment basis change $Q \mapsto LQ$ gives $G \mapsto LGL^\top$, $B \mapsto LB$, and leaves P unchanged. $\square$

2.2 Finite switch sets and the role of the constraints

Proposition 2.3. *On a compact interval, suppose the q_j are C^2 , that (1.2) holds, that all zeros of q_* are finitely many simple interior zeros, with no endpoint zeros, and that the Q_k span $\mathbb{R}^m$. For all sufficiently large M , the optimum in (1.1) is unique and consists of constant plateaus joined by linear ramps. Its value satisfies*

$$\delta(M) = \delta_0 + \frac{\delta_0^3 \Gamma}{3D} M^{-2} + o(M^{-3}).$$

If q_ is C^3 near the zeros, the full expansion (2.4)–(2.5) holds with finite sums.*

Proof. Root separation and sign persistence are automatic on a compact interval after shrinking the neighborhood of b^* . The balance equations define centers $c_k(a, b)$. At $a = 0$ the lower-moment map of their ramp profile has b -derivative $-G$. Expressing its local contributions as averages of primitives shows that this map is C^2 , including at $a = 0$. The implicit function theorem supplies an even C^2 multiplier $b(a)$ which makes all lower moments exactly zero. Thus $b(a) = b^* + va^2 + o(a^2)$, with v as above.

Lemma 2.2 proves exact support optimality. Equality forces $s = \text{sgn } q_b$ on every plateau and $s' = \sigma_k/a$ almost everywhere in every ramp, because $\sigma_k J < 0$ there. Continuity fixes all endpoint values. The support maximizer, and hence the constrained optimizer at the resulting budget, is unique. The target satisfies $S(a) = D - \Gamma a^2/3 + o(a^3)$ under C^2 regularity; its derivative is $O(a)$. Hence $M(a) = f/(aS(a))$ has negative derivative for small $a > 0$ and covers all sufficiently large budgets. The scalar inversion gives the stated second-order result. If q_* is C^3 near the switches, the finite version of (2.10) gives the fourth coefficient. $\square$

Remark 2.4. The sign of C_4 is not fixed by positivity of an underlying weight. The nonnegative quantity P has a more specific interpretation. For the fixed residual q_* , the unconstrained ramp support has fourth coefficient R ; imposing the lower moments gives $\Xi = R - P$. Thus the loss at this order is Pa^4 . This does not compare the original q_m problem with a different problem in which all lower constraints have simply been deleted.

3 Exact examples and a regularity obstruction

These examples have $X = [-1, 1]$, $m = 1$, $q_0 = 1$, and $b^* = 0$. They permit direct integration, independently of the coefficient formulas.

3.1 An asymmetric two-switch example

Let

$$q_1(x) = (x^2 - \tfrac{1}{4})(1 + \beta x), \quad |\beta| < 1, \quad f = \tfrac{1}{8}.$$

The zeros are $\pm 1/2$, and the sign has zero mean. For small a , exact mean balance puts the two centers at $c_{\pm} = d \pm 1/2$. Their individual balance equations coincide:

$$2d + 3\beta d^2 + \beta a^2 = 0.$$

The solution near zero is $d = -\beta a^2/2 - 3\beta^3 a^4/8 + O(a^6)$. Integration of the resulting profile gives

$$S(a) = \tfrac{1}{2} - \tfrac{2}{3}a^2 + \tfrac{\beta^2}{2}a^4 + O(a^6).$$

In particular,

$$\delta_0 = \tfrac{1}{4}, \quad C_2 = \tfrac{1}{48}, \quad C_4 = \tfrac{16/3 - \beta^2}{1024}. \quad (3.1)$$

As an exact arithmetic check, for $\beta = 1/4$,

$$G = \tfrac{256}{63}, \quad B = \tfrac{244}{189}, \quad R = \tfrac{134}{567}, \quad P = \tfrac{3721}{18144}, \quad \Xi = \tfrac{1}{32}, \quad C_4 = \tfrac{253}{49152}.$$

The cancellation producing Ξ is substantial enough that retaining the moment correction matters.

3.2 A negative fourth coefficient

Let $q_1(x) = x - 3x^3 + 9x^5$ and $f = 5/8$. Since $q_1 = x\{(3x^2 - 1/2)^2 + 3/4\}$, there is a single simple switch at zero. Odd symmetry gives the balanced center exactly. The supported moment is

$$S(a) = \tfrac{5}{2} - \tfrac{1}{3}a^2 + \tfrac{3}{10}a^4 - \tfrac{3}{7}a^6.$$

Inverting with $a = A/M$ gives

$$\delta(M) = \tfrac{1}{4} + \frac{1}{480M^2} - \frac{1}{15360M^4} + O(M^{-6}). \quad (3.2)$$

The underlying weight here is the positive constant 1, so it cannot enforce positivity of C_4 .

3.3 Why twice continuous differentiability is insufficient

Let $q_1(x) = x + \operatorname{sgn}(x)|x|^{5/2}$ and $f = 11/28$. This function is C^2 , with a simple zero at zero, but is not C^3 . The exact ramp moment is

$$S(a) = \tfrac{11}{7} - \tfrac{a^2}{3} - \tfrac{8a^{7/2}}{63}.$$

Consequently,

$$\delta(M) = \tfrac{1}{4} + \frac{7}{2112M^2} + \frac{1}{6336M^{7/2}} + O(M^{-4}). \quad (3.3)$$

Thus a finite coefficient in a remainder of the form $C_4 M^{-4} + o(M^{-4})$ after the second-order term does not exist. This is consistent with the $o(M^{-3})$ conclusion under C^2 .

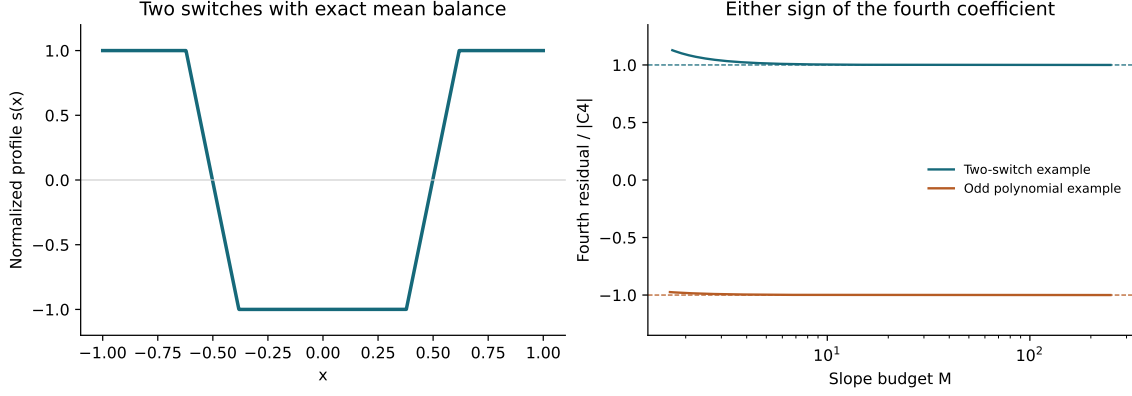


Figure 1: Left: the unique normalized optimizer in the two-switch example with $\beta = 1/4$ at ramp half-width $a = 0.12$, computed from its exact balance equation. Right: the scaled residual $M^4(\delta(M) - \delta_0 - C_2/M^2)/|C_4|$ for the first two examples, with the analytic limits $+1$ and -1 as dashed lines. The curves are high-precision evaluations of their explicit scalar equations, not interval certificates.

4 A positive theta-kernel family

Define, for $u \geq 0$,

$$\Phi(u) = \sum_{n \geq 1} \pi n^2 e^{5u} (2\pi n^2 e^{4u} - 3) \exp(-\pi n^2 e^{4u}), \quad (4.1)$$

$$F(t; \lambda, \mu, \nu) = \int_0^\infty \Phi(u) e^{\lambda u^2 + \mu u^4 + \nu u^6} \cos(2tu) \, du. \quad (4.2)$$

Every summand of Φ is positive on this domain. Double exponential decay permits parameter differentiation on compact sets, including complex parameter sets. Write $D_j = \partial_t^j F$. Then

$$\partial_\lambda D_j = -D_{j+2}/4, \quad \partial_\mu D_j = D_{j+4}/16, \quad \partial_\nu D_j = -D_{j+6}/64. \quad (4.3)$$

4.1 The cusp branch and the design problem

A validated zero of (D_0, D_1, D_2) at $\nu = 0$ has the rounded location

$$(t, \lambda, \mu) \simeq (41.4003413586842508894, -3.64569206160491909881, 8.33512049891860723771).$$

These displayed coordinates are explanatory approximations; the companion certificate specifies the actual enclosing box. At that zero,

$$\begin{aligned} 3.33563055 \cdot 10^{-13} &< D_3 < 3.33563057 \cdot 10^{-13}, \\ -9.71679534 \cdot 10^{-13} &< D_4 < -9.71679532 \cdot 10^{-13}. \end{aligned}$$

In particular the control determinant for the equations $D_0 = D_1 = 0$ is $D_3 D_4 / 64 \neq 0$, and $D_3 \neq 0$ gives the usual local cusp geometry of the zero set of F . A computer-assisted continuation defines a single analytic branch

$$Q(\nu) = (\tau(\nu), \lambda(\nu), \mu(\nu)), \quad \nu \in I = [-1/64, 0].$$

Shared parameter faces are joined by root containment in the next uniqueness box, not merely by overlap of the surrounding boxes. The validated continuation methods are used to establish the particular branch and bounds here; cusp validation as a general method is established in the literature [5].

Differentiating the three cusp equations gives the useful identities

$$\begin{aligned}\mu' &= D_6/(4D_4), & \lambda' &= (4D_5\mu' - D_7)/(16D_3), \\ \tau' &= (D_8/64 + D_4\lambda'/4 - D_6\mu'/16)/D_3.\end{aligned}\tag{4.4}$$

Set

$$w_\nu(u) = \Phi(u)e^{\lambda(\nu)u^2 + \mu(\nu)u^4 + \nu u^6}, \quad p_{j,\nu}(u) = (2u)^j \cos(2\tau(\nu)u + j\pi/2),$$

and let $q_{j,\nu} = w_\nu p_{j,\nu}$, $0 \leq j \leq 3$. In (1.1) take $m = 3$ and $f_\nu = D_3(Q(\nu), \nu) > 0$, and denote its value by $\delta_\nu(M)$. All coefficients in Section 2 now depend on ν .

If g is feasible, multiplying the kernel by $1 - g$ makes the first four t -derivatives, of orders 0, 1, 2, 3, vanish at the prescribed point. If $\|g\|_\infty < 1$, the modified kernel is still positive. The size $\delta_\nu(M)$ is therefore the least amplitude of such a constrained change, with its roughness controlled by M . This interpretation concerns a separate profile for each ν .

Theorem 4.1 (Theta value enclosure). *For every $\nu \in I$, the sign certificate and hypotheses of Theorem 2.1 hold for the kernels just defined. For every $M \geq M_0 = 2 \cdot 10^{-5}$, the optimum is attained and*

$$9.17 \cdot 10^{-10} < \delta_\nu(M) < 9.181 \cdot 10^{-10} < 1.$$

With $P_{4,\nu}(M) = \delta_{0,\nu} + C_{2,\nu}M^{-2} + C_{4,\nu}M^{-4}$,

$$-\frac{1.02 \cdot 10^{-53}}{M^6} < \delta_\nu(M) - P_{4,\nu}(M) < \frac{1.64 \cdot 10^{-53}}{M^6}.\tag{4.5}$$

Furthermore, throughout I ,

$$2.48374879 \cdot 10^{-39} < C_{4,\nu} < 2.49203006 \cdot 10^{-39},\tag{4.6}$$

$$2.7 \cdot 10^{-11} < \delta'_{0,\nu} < 3.0 \cdot 10^{-11},$$

$$7.4 \cdot 10^{-26} < C'_{2,\nu} < 8.7 \cdot 10^{-26},$$

$$1.9 \cdot 10^{-40} < C'_{4,\nu} < 5.3 \cdot 10^{-40}.\tag{4.7}$$

Proof structure and computational evidence. The branch construction, sign equations, invertible moment Jacobian, simple residual zeros, and derivative bounds are established on 32 closed driver cells

$$I_i = [-(i+1)/2048, -i/2048], \quad 0 \leq i < 32.$$

The moment Jacobian is $-G$; its nonsingularity and the convexity of $\int |q_b|$ identify the local dual solution as the unique global minimizer. For clarity, a second minimizer would make the objective constant on their joining segment, contradicting its positive definite Hessian near the certified minimizer.

The residual without its positive weight has the form

$$r_b(u) = (8u^3 + 2b_1u) \sin(2\tau u) + (4b_2u^2 - b_0) \cos(2\tau u).\tag{4.8}$$

On $[0, 1]$, the evidence isolates 28 simple zeros and covers their complement by sign enclosures. On $u \geq 1$, put $T = ((8u^3 + 2b_1u)^2 + (4b_2u^2 - b_0)^2)^{1/2}$ and $\phi = 2\tau u + \arctan((4b_2u^2 - b_0)/(8u^3 + 2b_1u))$. The uniform bounds

$$T > 7u^3, \quad |T'| < 25u^2, \quad 81 < \phi' < 85$$

give separated, simple zeros, with spacing greater than $3/85$ and $|r'_b| > 567u^3$ at each zero. They also bound root motion with b , since $|\partial_{b_j} z| \leq 1/(81z)$ for the tail zeros.

The weight decay verifies the stronger, weighted hypothesis (2.3), rather than only unweighted convergence. For $H = 1/64$ and $\rho \leq 1/1000$, suitable uniform constants give

$$\begin{aligned} E_k &\leq C(1 + z_k + \rho)^{18} e^{21(z_k + \rho) + 9(z_k + \rho)^4 - \pi e^{4(z_k - \rho)}}, \\ L_k &= e^{-4(z_k + \rho)^2 - H(z_k + \rho)^6 - \pi e^{4(z_k + \rho)}}. \end{aligned}$$

The ratio E_k^3/L_k^2 is bounded by a polynomial times

$$\exp\{63(z_k + \rho) + 27(z_k + \rho)^4 + 8(z_k + \rho)^2 + 2H(z_k + \rho)^6 - \pi c_\rho e^{4z_k}\}, \quad c_\rho = 3e^{-4\rho} - 2e^{4\rho} > 0.97.$$

The separated zeros and this double exponential decay prove (2.3). Finite initial neighborhoods are covered directly. Theorem 2.1 now applies.

The explicit remainder requires additional estimates and does not follow just from that theorem. Appendix A constructs finite ramp profiles with exact moments and an accompanying support bound, uniformly on all 32 cells. Scalar comparison, without continuity of the number of active ramps, proves (4.5) and feasibility for every stated M . Differentiated coefficient identities and interval enclosures give (4.6)–(4.7). The exact input-output map and replay commands appear in Appendix C; those finite computations are part of this computer-assisted theorem. $\square$

At $M = M_0$, the absolute error in (4.5) lies between $-1.59375 \cdot 10^{-25}$ and $2.5625 \cdot 10^{-25}$. These figures measure the truncation error once the coefficients are known; rounding the coefficients separately introduces additional error. In particular the broad uniform ranges above are not substitutes for pointwise coefficient enclosures.

5 Transport and strict ordering of the optimal value

The coefficient derivatives in (4.7) and a uniform remainder suffice to compare well-separated parameters. They do not by themselves compare the actual values at arbitrarily close parameters. We use a change of scale which preserves all the moment equations. Weighted composition operators on Lipschitz spaces are an established framework [4]; the result below is a sufficient differential condition adapted to these moments and to a specified amplitude-to-slope ratio.

Theorem 5.1 (Moment-preserving transport). *Suppose $w_\nu(u) > 0$ is C^2 in u , is C^1 in ν , and has continuous mixed derivative $\partial_u \partial_\nu \log w_\nu$. Suppose $f_\nu > 0$ and $\tau(\nu) > 0$ are C^1 . Assume that the following kernels are integrable on the half-line:*

$$q_{j,\nu}(u) = w_\nu(u)(2u)^j \cos(2\tau(\nu)u + j\pi/2), \quad 0 \leq j \leq m.$$

Let $\kappa = -\tau'/\tau \geq 0$, $L_\nu = \partial_u \log w_\nu$, and

$$r_\nu(u) = -f'_\nu/f_\nu + (m+1)\kappa + \partial_\nu \log w_\nu(u) + \kappa u L_\nu(u).$$

If for constants $a_0, c > 0$

$$r_\nu + \kappa + a_0 |\partial_u r_\nu| \leq -c \tag{5.1}$$

throughout a parameter interval and the whole half-line, then for $\nu_1 < \nu_2$, $d = \nu_2 - \nu_1$, $\eta = \tau(\nu_1)/\tau(\nu_2)$, the transformation

$$(\mathcal{T}g)(u) = T(u)g(\eta u), \quad T(u) = \frac{f_{\nu_1}}{f_{\nu_2}} \eta^{m+1} \frac{w_{\nu_2}(\eta u)}{w_{\nu_1}(u)} \tag{5.2}$$

preserves feasibility of the moment equations from ν_2 to ν_1 . If $\|g\|_\infty \leq a_0 M$, $\text{Lip}(g) \leq M$, then

$$\|\mathcal{T}g\|_\infty \leq e^{-cd}\eta^{-1}\|g\|_\infty, \quad \text{Lip}(\mathcal{T}g) \leq e^{-cd}M.$$

Consequently, whenever the optimum at ν_2 is attained with amplitude at most $a_0 M$,

$$\delta_{\nu_2}(M) \geq \eta e^{cd} \delta_{\nu_1}(M). \quad (5.3)$$

Proof. Substitution $x = \eta u$ and $p_{j,\nu_2}(\eta u) = \eta^j p_{j,\nu_1}(u)$ give the exact identity

$$\int q_{j,\nu_1} \mathcal{T}g = \frac{f_{\nu_1}}{f_{\nu_2}} \eta^{m-j} \int q_{j,\nu_2} g.$$

This proves all moments, including the target at $j = m$.

For the norm estimate, set $\nu(t) = \nu_2 - t$ and $\eta(t) = \tau(\nu(t))/\tau(\nu_2)$, and use (5.2) with this moving first endpoint. Define $\mathcal{D} = \partial_t - \kappa u \partial_u$. Direct logarithmic differentiation gives

$$\mathcal{D}T = rT, \quad \eta' = \kappa\eta, \quad \mathcal{D}(T_u) = (r + \kappa)T_u + r_u T.$$

The positive commutator term $+\kappa T_u$ is essential. Since $\eta \geq 1$, $T > 0$, the quantity $W = \eta T + a_0 |T_u|$ satisfies $T \leq W$, and, along each characteristic,

$$\mathcal{D}W \leq (r + \kappa)W + a_0 |r_u| T \leq -cW.$$

The inequality for the absolute value is valid almost everywhere; alternatively it follows by regularization. Initially $T = 1, T_u = 0$, so $W(0, u) = 1$. Every finite endpoint characteristic has a finite initial point. Thus no uniform boundedness of r at infinity is needed, and

$$\eta T + a_0 |T_u| \leq e^{-ct}, \quad T \leq e^{-ct}/\eta.$$

The product and chain rules for Lipschitz functions prove the two norm bounds. Applying them to an optimizer at ν_2 provides a feasible competitor at ν_1 and proves (5.3). $\square$

Corollary 5.2 (Strict order on the theta branch). *For the family in Section 4, for every $M \geq 2 \cdot 10^{-5}$ and every $-1/64 \leq \nu_1 < \nu_2 \leq 0$,*

$$\delta_{\nu_2}(M) \geq \frac{\tau(\nu_1)}{\tau(\nu_2)} e^{(\nu_2 - \nu_1)/50} \delta_{\nu_1}(M) > \delta_{\nu_1}(M), \quad (5.4)$$

$$\begin{aligned} \log \delta_{\nu_2}(M) - \log \delta_{\nu_1}(M) &> 0.02259(\nu_2 - \nu_1), \\ \delta_{\nu_2}(M) - \delta_{\nu_1}(M) &> 2.07 \cdot 10^{-11}(\nu_2 - \nu_1). \end{aligned} \quad (5.5)$$

These inequalities do not require differentiability or uniqueness of the finite-budget optimizer.

Proof. The validated branch bounds include

$$.00259 < \kappa < .00260, \quad .0437 < f'_\nu/f_\nu < .0451, \quad .3260 < \lambda' < .3264, \quad .2939 < \mu' < .2942.$$

With $a_0 = 2^{-14}$, Appendix B proves (5.1) with $c = 1/50$ on the whole half-line. Theorem 4.1 gives $\delta_\nu(M)/M < 9.181 \cdot 10^{-10}/(2 \cdot 10^{-5}) < a_0$, so Theorem 5.1 applies. Integrating the strict lower bound for κ proves the logarithmic bound. Finally $e^x - 1 \geq x$ and $9.17 \cdot 10^{-10} \cdot .02259 > 2.07 \cdot 10^{-11}$ give (5.5). $\square$

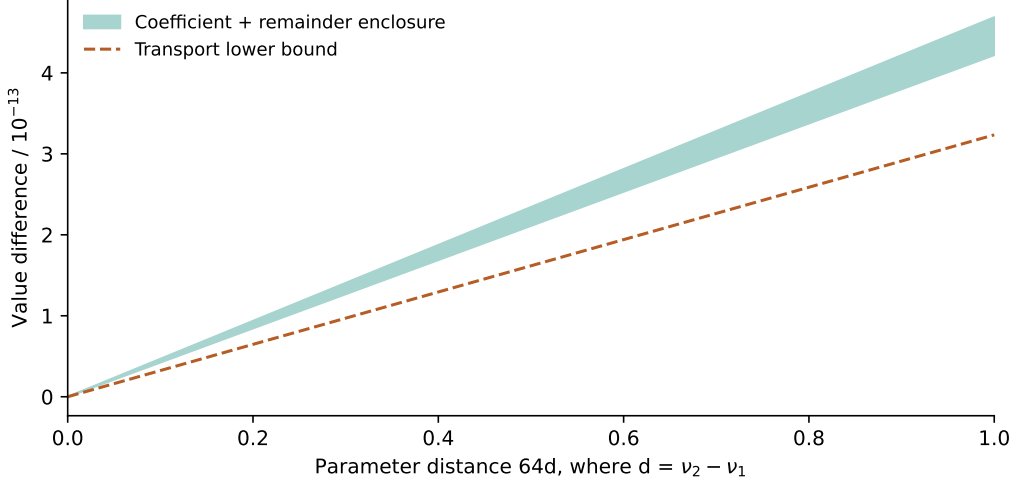


Figure 2: Guaranteed bounds on the change of the optimal value at $M = 2 \cdot 10^{-5}$, for any two driver parameters at distance d . The shaded region is the coefficient-plus-remainder enclosure derived in this section. The dashed line is the weaker but strictly positive transport bound $2.07 \cdot 10^{-11}d$. This plot displays bounds, not numerically computed finite-budget optimizers. At $d = 0$ the exact difference is zero.

For a fixed nonzero separation, the coefficient bounds can give a sharper two-sided estimate. Integrating (4.7) and subtracting the two errors in (4.5) gives, with $d = \nu_2 - \nu_1$,

$$\mathcal{L}(M)d - \frac{2.66 \cdot 10^{-53}}{M^6} < \delta_{\nu_2}(M) - \delta_{\nu_1}(M) < \mathcal{U}(M)d + \frac{2.66 \cdot 10^{-53}}{M^6},$$

where

$$\begin{aligned} \mathcal{L}(M) &= 2.7 \cdot 10^{-11} + 7.4 \cdot 10^{-26}M^{-2} + 1.9 \cdot 10^{-40}M^{-4}, \\ \mathcal{U}(M) &= 3.0 \cdot 10^{-11} + 8.7 \cdot 10^{-26}M^{-2} + 5.3 \cdot 10^{-40}M^{-4}. \end{aligned}$$

The transport estimate remains strictly positive however small $d > 0$ is; the fixed error allowance in the two-sided estimate cannot do this.

6 Scope, reproducibility, and questions for review

The resulting picture is that smoothing a sign-optimal moment design has a computable leading amplitude cost and a fourth-order correction which records the coupling to the lower moments. In the theta example, this description is effective at stated budgets, and the least required amplitude increases along the chosen cusp parameter interval. This is the concrete optimization conclusion of the paper.

The finite-switch result identifies a unique optimizing profile for sufficiently large M . The countably infinite result identifies the value asymptotics and proves attainment; it does not establish uniqueness or differentiability of the finite- M optimizer. The theta error bound is specific to $\nu \in [-1/64, 0]$ and $M \geq 2 \cdot 10^{-5}$. It is not a uniform result on the longer exploratory cusp arc.

The target f_ν , the kernels, and the optimizing profile all vary with ν . Strict comparison concerns these explicitly defined, different optimization problems. It is not a comparison with a common target or a common profile. The expansion with an $O(M^{-6})$ error does not identify a coefficient of M^{-6} . Nor does the paper prove monotonicity of a normalized fourth-order remainder at arbitrarily close parameters.

The moment construction gives a zero of order at least four for the modified integral at its prescribed point. Exactly fourth order, a nondegenerate unfolding of the modified family, and global root counts require further conditions. No conclusion about the Riemann hypothesis, a de Bruijn–Newman constant, a physical system, or an AI training procedure is drawn.

The accompanying source and evidence index identify which claims are analytic, which use outward interval bounds, and which numerical plots are diagnostic. An interval certificate is a finite collection of enclosures used with stated analytic arguments; it is not a certification of the manuscript as a whole. The checks can expose inconsistencies and reproduce enclosures, but are not a machine-checked proof of the analytic arguments.

The main requests to an independent reviewer are: the weighted Taylor interchanges and finite moment repair in Theorem 2.1; the effective primal and dual estimates behind Theorem 4.1; and the characteristic inequality used in Theorem 5.1. Questions of novelty relative to bounded Lipschitz optimization are also open to review.

Use of AI-assisted tools. AI assistants were used in exploratory calculations, code development, proof drafting, and internal checking. Exact algebra and interval computations are reported separately from those assistants’ assessments. Independent expert review is being sought. The author is responsible for the submitted claims.

A Effective support bounds and exact feasibility

This appendix records the analytic estimates used by the finite interval calculations. All constants below are evaluated on whole parameter and root boxes. Point evaluations at approximate roots do not suffice.

A.1 Local estimates

Put $a_0 = 2^{-14}$, $\epsilon = 2^{-12}$ and $b(a) = b^* + va^2$. At an original residual zero z_k , start with the center $c_k = z_k + \kappa_k a^2$, with κ_k from (2.7). Write $K = |\kappa_k|$, $Q_{j,\ell,k} = \sup_{N_k} |q_j^{(\ell)}|$, and $R_{\ell,k} = \sup_{N_k} |q_*^{(\ell)}|$. In the following formulas, unlabelled derivatives are evaluated at z_k . The moment change at this switch is exactly

$$\Delta_{j,k} = -2\sigma_k \mathbb{E} \int_0^{\kappa_k a^2 + ay} q_j(z_k + x) dx.$$

Taylor’s formula gives

$$|\Delta_{j,k} - a^2(-2\sigma_k q_j \kappa_k - \sigma_k q_j'/3)| \leq H_{j,k} a^4,$$

where

$$\begin{aligned} H_{j,k} &= |q_j'| K^2 + \frac{|q_j''|}{3} (K + K^3 a_0^2) \\ &\quad + \frac{Q_{j,3,k}}{12} (1/5 + 2K^2 a_0^2 + K^4 a_0^4). \end{aligned}$$

For the residual, whose value at its zero is exactly zero,

$$\Delta_{*,k} = -\sigma_k r_k a^2/3 - \sigma_k (r_k \kappa_k^2 + t_k \kappa_k/3 + u_k/60) a^4 + \varepsilon_k, \quad |\varepsilon_k| \leq K_{*,k} a^6,$$

with

$$\begin{aligned} K_{*,k} &= |t_k| K^3/3 + |u_k| (2K^2 + K^4 a_0^2)/12 \\ &\quad + |q_*^{(4)}| (K + (10/3) K^3 a_0^2 + K^5 a_0^4)/60 \\ &\quad + R_{5,k} (1/7 + 3K^2 a_0^2 + 5K^4 a_0^4 + K^6 a_0^6)/360. \end{aligned}$$

These coefficients come from signed averages of powers, including $\mathbb{E}(\kappa a^2 + ay)^5 = \kappa a^6 + (10/3)\kappa^3 a^8 + \kappa^5 a^{10}$. Bounding that odd average by an absolute fifth moment would lose the required order.

For $F_k(c) = \mathbb{E}q_{b(a)}(c + ay)$, the quadratic coefficient at c_k vanishes because $r_k \kappa_k + t_k/6 - Q_k^\top v = 0$. Thus $|F_k(c_k)| \leq h_k a^4$, where

$$\begin{aligned} h_k &= |t_k|K^2/2 + |u_k|(K + K^3 a_0^2)/6 \\ &\quad + R_{4,k}(1/5 + 2K^2 a_0^2 + K^4 a_0^4)/24 \\ &\quad + \sum_{j < 3} |v_j| \{ |q'_j|K + Q_{j,2,k}(1/3 + K^2 a_0^2)/2 \}. \end{aligned}$$

If the signed derivative $\sigma_k F'_k \geq m_k > 0$ on the relevant center interval, the true balanced center is within $h_k a^4/m_k$. The local objective has second derivative at most $-2m_k$. Its possible increase after balancing is consequently at most $h_k^2 a^8/m_k$. Containment of both ramps is checked before this estimate is applied.

Move the first three centers by $a^4 y_k$, with $|y_k| \leq W_k$, $W = (2^{26}, 2^{26}, 2^{24})$. For $J_{jk} = -2\sigma_k q_j(z_k)$, let C be the dyadic preconditioner recorded for the cell. The Jacobian deviation is bounded by

$$\Delta J_{jk} \leq 2\{Q_{j,1,k}(K_k a_0^2 + W_k a_0^4) + Q_{j,2,k} a_0^2/6\}.$$

In the norm $\max |y_k|/W_k$, the map subtracting C times the lower moments divided by a^4 has the following uniform bounds:

$$\text{derivative rows} < (.005731, .007842, .004703), \quad \text{forcing} < (.328781, .448496, .267009).$$

Each corresponding sum is less than one. The contraction theorem gives exactly vanishing lower moments within this center construction. The same estimates imply that C is nonsingular. With $\mathcal{L}_k = \sup |q'_{b(a)}|$, the repair costs at most

$$a^8 \sum_{k=1}^3 (2h_k W_k + \mathcal{L}_k W_k^2).$$

This estimate is made for $q_{b(a)}$, whose center derivative is of order a^4 . The repaired target equals its q_3 target because the lower moments vanish.

A.2 A changing finite prefix and its entire omitted tail

Always retain the 28 zeros below 1. Retain a tail zero $z_k > 1$ precisely when $ae^{4z_k} \leq \epsilon$. This gives a finite prefix for every $a > 0$. After the last active ramp, extend its final plateau to infinity. The first inactive zero is separated from the finite interval by the validated positive gap $[1, 1.001]$.

Uniformly on the driver cells, $|v| < (5, 4, 180)$ and the expanded trial multiplier box satisfies $|b| < (.001, .1, .005)$. On each active tail neighborhood $[z_k - 3a, z_k + 3a]$, the raw phase estimates give $|\kappa_k| < 46e^{4z_k}$, signed raw trial derivative greater than $550z_k^3$, and raw absolute value less than $2800az_k^3$. Also $\frac{1}{2} < w(x)/w(z_k) < 2$, since the logarithmic variation is less than $408\epsilon < 1/2$. Combining the weight derivative with these estimates gives $\sigma_k q'_{b(a)} > 200w(z_k)z_k^3$. Thus balancing is valid on active tail neighborhoods as well.

For $u \geq 1$, the weight satisfies

$$w_\nu(u) \leq 88e^{9u+9u^4-\pi e^{4u}}.$$

Writing $P_\nu = \lambda u^2 + \mu u^4 + \nu u^6$ and $H = 1/64$,

$$\begin{aligned} |P'_\nu| &\leq 44u^3 + 6Hu^5, & |P''_\nu| &\leq 116u^2 + 30Hu^4, \\ |P'''_\nu| &\leq 216u + 120Hu^3, & |P^{(4)}_\nu| &\leq 216 + 360Hu^2, & |P^{(5)}_\nu| &\leq 720Hu. \end{aligned}$$

The inequalities $e^{4u} > 50u^3$ and $u^5 e^{-4u} < 1/40$ imply $|P^{(j)}_\nu| < e^{4ju}$, $1 \leq j \leq 5$. The theta derivatives follow from

$$P_0(x) = -3 + 2x, \quad P_{\ell+1}(x) = (5 - 4x)P_\ell(x) + 4xP'_\ell(x),$$

applied term by term to Φ ; product rules and exponential derivative polynomials give the needed five weight derivatives. This retains the sextic terms in local calculations and discards their negative exponent only when taking an upper tail bound.

For $0 \leq p \leq 3$, $d \in \{12, 16, 24\}$, the corresponding root sum and integral are bounded by

$$S_d = \frac{88e^{18+d-\pi e^4}}{1 - 50^{-4}}, \quad I_d = \frac{88}{500}e^{18+d-\pi e^4}.$$

Indeed the logarithmic derivative of the envelope is below -500 ; root spacing exceeds $3/85$, giving a successive term ratio below 50^{-4} . For every nonnegative sequence t_k , inactive roots satisfy

$$\sum_{ae^{4z_k} > \epsilon} t_k \leq (a/\epsilon)^s \sum_k t_k e^{4sz_k} \quad (s > 0).$$

This supplies the missing powers of a ; a fixed cutoff would not. For example the inactive quadratic and fourth coefficients are paid for by $300\epsilon^{-4}S_{16}$ and $2071000\epsilon^{-2}S_{16}$, respectively, at sixth order. The lower-moment coefficient uses $188\epsilon^{-2}S_{12}$. Cutting the final integral at the first inactive zero minus $3a$ gives the additional bounds

$$\frac{18\epsilon^{-6}I_{24}}{1 - 72a_0}, \quad \frac{2^{j+1}\epsilon^{-4}I_{16}}{1 - 48a_0}$$

for the normalized objective and lower-moment tails. Active tail derivatives use the same envelope with the local formulas above. The evidence records these sums, signs, and containment margins.

Combining the finite part, the active tail, and the inactive tail yields, for every fixed $(\nu, a) \in I \times (0, a_0]$,

$$H_{b(a)}(a) \leq D - \Gamma a^2/3 + \Xi a^4 + K_d a^6, \tag{A.1}$$

$$|S(a) - (D - \Gamma a^2/3 + \Xi a^4)| \leq K_p a^6, \tag{A.2}$$

where $K_d < 655000$ and $K_p < 4.746 \cdot 10^6$ suffice uniformly. During each three-center contraction, a and hence the prefix are fixed. Continuity of the prefix as a function of a is not used.

A.3 Scalar feasibility and inversion

Set $U = 9.181 \cdot 10^{-10}$ and define

$$F_M(A) = DA - \Gamma A^3/(3M^2) + \Xi A^5/M^4, \quad J_M(A) = F_M(A) - K_p A^7/M^6.$$

On every driver cell, interval evaluation proves, for all $M \geq M_0$,

$$J_M(\delta_0) < f < J_M(U), \quad J'_M > 0 \text{ on } [\delta_0, U], \quad U/M_0 < a_0.$$

The upper endpoint's normalized margin exceeds $1.7683 \cdot 10^{-15}$. Let A_+ be the scalar root $J_M(A_+) = f$. At $a = A_+/M$, (A.2) gives $S(a) \geq f/A_+$. Multiplication of the repaired profile by $A = f/S(a) \leq A_+$ gives exact target and slope $A/a \leq M$. This proves feasibility at every budget despite changes in the prefix. Attainment follows as in Theorem 2.1.

For the optimizer, (A.1) gives $F_M(\delta(M)) \geq f - K_d U^7/M^6$. The scalar upper root gives the corresponding upper comparison. To bound the residual of P_4 , put

$$w = (\delta_0/M)^2, \quad t = \Gamma/(3D), \quad e = \Xi/D, \quad p = 1 + tw + (3t^2 - e)w^2.$$

Then $F_M(P_4)/f - 1 = p - twp^3 + ew^2p^5 - 1$. This is a polynomial of degree twelve whose coefficients through degree two vanish exactly. Bounding all remaining coefficients on $0 \leq w \leq (\delta_0/M_0)^2$ gives $|F_M(P_4) - f| \leq K_{\text{poly}}/M^6$. Also $P_4 \in [\delta_0, U]$ and

$$F'_M \geq d_{\min} = D_{\text{lo}} - \Gamma_{\text{hi}} U^2/M_0^2 > 0.$$

The inverse comparison constants are

$$K_- = \frac{K_{\text{poly}} + K_d U^7}{d_{\min}} < 1.001 \cdot 10^{-53}, \quad K_+ = \frac{K_{\text{poly}} + K_p U^7}{d_{\min}} < 1.620 \cdot 10^{-53}.$$

Rounding these bounds outwards gives (4.5).

B Verifying the transport condition on an unbounded domain

The uniform caps used here are

$$\lambda \in (-3.651, -3.645), \quad \mu \in (8.330, 8.336), \quad \nu \in [-1/64, 0],$$

together with the derivative caps in the proof of Corollary 5.2. For $\Phi = \sum \phi_n$, put $x = \pi e^{4u}$ and, for $n \geq 2$,

$$R_n = \frac{\phi_n}{\phi_1} = n^2 \frac{2n^2 x - 3}{2x - 3} e^{-(n^2-1)x}, \quad \ell_n = 9 + \frac{12}{2n^2 x - 3} - 4n^2 x.$$

Then $\ell'_n = -16n^2 x - 96n^2 x/(2n^2 x - 3)^2$. For $R = \sum_{n \geq 2} R_n$,

$$L_\Phi = \ell_1 + \frac{R'}{1+R}, \quad L'_\Phi = \ell'_1 + \frac{R''}{1+R} - \left(\frac{R'}{1+R} \right)^2. \quad (\text{B.1})$$

On $x \geq 3$,

$$R_n \leq 2n^4 e^{-(n^2-1)x}, \quad |R'_n| \leq 10n^6 x e^{-(n^2-1)x}, \quad |R''_n| \leq 100n^8 x^2 e^{-(n^2-1)x}.$$

These follow by differentiating the displayed ratio; its logarithmic derivative has absolute value at most $5n^2 x$, and the difference of the logarithmic second derivatives has absolute value at most $20n^2 x$. The summable majorants also justify the differentiations.

For $0 \leq u \leq 1$, include $n = 2, \dots, 12$ explicitly. The remaining errors in R, R', R'' are bounded by

$$4 \cdot 13^4 e^{-504}, \quad 60 \cdot 13^6 e^{-504}, \quad 1800 \cdot 13^8 e^{-504},$$

because consecutive majorants have ratio below $1/2$. Arb evaluation of (B.1) on each of the 1024 whole intervals $[i/1024, (i+1)/1024]$, using all parameter caps simultaneously, then bounds

$$\begin{aligned} L_\nu &= L_\Phi + 2\lambda u + 4\mu u^3 + 6\nu u^5, \\ r &= -f'/f + 4\kappa + \lambda' u^2 + \mu' u^4 + u^6 + \kappa u L_\nu, \\ r_u &= 2\lambda' u + 4\mu' u^3 + 6u^5 + \kappa(L_\nu + u L'_\nu). \end{aligned}$$

The largest recorded upper endpoint of $r + \kappa + a_0|r_u|$ is less than -0.02170697 ; a separate outward rational reconstruction proves the simpler bound $< -0.0217 < -1/50$ on every cell.

For $u \geq 1$, $x > 150$ and

$$R < 64e^{-450}, \quad |R'| < 192000e^{-450}, \quad |R''| < 1152000000e^{-450}.$$

Each is less than $1/4$, and $R' < 0$. Equation (B.1) therefore gives

$$L_\Phi \leq 10 - 4x, \quad |L_\Phi| \leq 4x + 1, \quad |L'_\Phi| \leq 16x + 2.$$

For $H = 1/64$,

$$\begin{aligned} L_\nu &\leq 10 - 4x + 36u^3 < 0, \\ |L_\nu| &\leq 4x + 1 + 8u + 36u^3 + 6Hu^5, \\ |L'_\nu| &\leq 16x + 10 + 108u^2 + 30Hu^4. \end{aligned}$$

Set $k_- = .00259$, $k_+ = .00260$, $l = .3264$, $m = .2942$, $F = .0437$, and $\bar{k} = k_- - 5a_0k_+ > 0$. Using the sign of L_ν in the term without absolute values,

$$r + \kappa + a_0|r_u| \leq P_6(u) - 4\bar{k}\pi ue^{4u} =: Q(u),$$

where, in increasing degree, the coefficients of P_6 are

$$\begin{aligned} &-F + 5k_+ + a_0k_+, \quad 10k_- + a_0(2l + 18k_+), \quad l, \\ &a_0(4m + 144k_+), \quad m + 36k_-, \quad a_0(6 + 36Hk_+), \quad 1. \end{aligned}$$

All positive-degree coefficients are positive. With $S = \sum_{j=1}^6 jP_j$, $P'_6(u) \leq Su^5$. The inequality $(1 + 4u)e^{4u} > 270u^5$ for $u \geq 1$ follows on writing $z = u - 1$, using $e^4 > 109/2$ and the first six Taylor terms for e^{4z} . The difference polynomial has initial coefficients $5/2, -42, 352$, forming a positive quadratic, and all higher coefficients positive. Exact rational arithmetic then gives

$$\begin{aligned} Q'(u) &< (S - 4\bar{k} \cdot 3 \cdot 270)u^5 < 0, \\ Q(1) &< \sum P_j - 4\bar{k} \cdot 170 \\ &= -0.051116612396240234375 < -1/50. \end{aligned}$$

This proves the required inequality on the entire tail.

C Evidence, branch identification, and replay

The supplement preserves the numbered research packages as a fixed snapshot. Their identifiers describe provenance, not additional unpublished theorems which a reader must accept without a proof. The analytic arguments used in this paper are given above; detailed local numerical witnesses and coefficient differentiation formulas are in the indicated files.

Claim	Evidence ages	pack-	Principal material
General expansion	R21–R23		Finite and countable-switch proofs, exact algebra, examples
Cusp identity	R01, R03, R09, R29		Integral root box, connected uniqueness tubes, positive majorants, 32 driver cells
Theta coefficients	R24, R28, R29		Full root sums, weighted tails, implicit differentiation
Effective remainder	R25–R27, R30		Local jets, finite repair, moving prefix, scalar inversion
Actual-value order	R31		Moment identity, 1024 spatial cells, whole-tail comparison

For reproducibility, the supplement includes the source, data, and dependency closure of these packages. Its index maps each identifier to a directory, manifest, proof, producer, and checker. The following commands use paths relative to the supplement root:

```
python -B research/finite_slope_optimum/audit_snapshot.py
python -B research/finite_slope_fourth_order/audit_snapshot.py
python -B research/infinite_slope_fourth_order/audit_snapshot.py --with-arb
python -B research/cusp_order_transport/audit_snapshot.py --with-arb
```

The last command recursively checks the later theta dependencies. It regenerates the 1024 spatial interval enclosures, reconstructs the 32 driver-cell chain, and checks arithmetic and recorded prerequisites. The accompanying report distinguishes freshly evaluated transcendental enclosures from frozen prerequisite inputs. The implementation uses Python, python-flint/Arb [6, 7], exact rational arithmetic, and mpmath for non-rigorous diagnostics. Pinned versions and input hashes are supplied.

Some details concerning identity and differentiation deserve emphasis. The initial cusp is enclosed by an integral-based contraction, with $D_3D_4 \neq 0$. In the original continuation, the scaled contraction bound is below 0.502 and the forcing-plus-contraction bound is below 0.560. At adjacent faces a smaller root enclosure lies inside the next full uniqueness box. On the short interval used here, refined anchors and degree-five parameter expansions with sixth-order remainder remain inside those already identified tubes. Thus a collection of unrelated zeros is not substituted for a branch.

The lower sign-moment equations have derivative $-G$. Their implicit solution $b^*(\nu)$, and the simple residual zeros, are differentiated together. For example,

$$\delta'_0 = (f'D - fD')/D^2, \quad C'_2 = C_2(3f'/f - 4D'/D + \Gamma'/\Gamma).$$

The $b^{*'} contribution to D' vanishes by the sign moments; it does not vanish from Γ' . Root-sum derivatives retain root motion. Their tail bounds are justified by separated zeros, implicit derivative bounds, and double exponential majorants of the same type as (2.3). The package R29 gives the termwise formulas and enclosing intervals used for (4.7).$

The final transport audit records 31 exact algebra groups, 19,017 rational decisions, and 1024 closed spatial cells. Its independent high-precision integral and composition diagnostics are useful consistency checks, but are not interval proofs. The rational checker uses stated transcendental enclosures as inputs; it is not a second independent rigorous integration engine. The four replay entry points passed on 23 September 2026, and 1,334 entries in 32 pre-existing manifests remained unchanged. These counts establish the tested snapshot and the coverage of that replay. They do not establish that every line of analytic reasoning is error-free.

For the public supplement, local machine paths in bookkeeping records are replaced by neutral placeholders. All dependent checksum references are updated consistently, and the complete transformation map is supplied. No mathematical formula or numerical enclosure is changed by that transformation. The public copy is replayed separately; its manifest identities are therefore distinguished from the original local snapshot.

References

- [1] J. Lellmann, D. A. Lorenz, C. Schönlieb, and T. Valkonen, *Imaging with Kantorovich–Rubinstein discrepancy*, SIAM Journal on Imaging Sciences 7 (2014), 2833–2859. doi:10.1137/140975528. Preprint: <https://arxiv.org/abs/1407.0221>.
- [2] S. C. Hille and E. S. Theewis, *Explicit expressions and computational methods for the Fortet–Mourier distance of positive measures to finite weighted sums of Dirac measures*, Journal of Approximation Theory 294 (2023), 105947. Preprint: <https://arxiv.org/abs/2206.12234>.
- [3] S. C. Hille and E. S. Theewis, *Norming and dense sets of extreme points of the unit ball in spaces of bounded Lipschitz functions*, Journal of Mathematical Analysis and Applications 536 (2024), 128200. Preprint: <https://arxiv.org/abs/2305.08500>.
- [4] A. Abbar, C. Coine, and C. Petitjean, *A pre-adjoint approach on weighted composition operators between spaces of Lipschitz functions*, preprint, 2023. <https://arxiv.org/abs/2304.12173>.
- [5] J.-P. Lessard and A. Pugliese, *Cusp bifurcations: numerical detection via two-parameter continuation and computer-assisted proofs of existence*, Linear Algebra and its Applications 728 (2026), 26–46. doi:10.1016/j.laa.2025.08.015. Preprint: <https://arxiv.org/abs/2404.00535>.
- [6] F. Johansson, *Arb: Efficient arbitrary-precision midpoint-radius interval arithmetic*, preprint, 2016. <https://arxiv.org/abs/1611.02831>.
- [7] F. Johansson, *Numerical integration in arbitrary-precision ball arithmetic*, preprint, 2018. <https://arxiv.org/abs/1802.07942>.